MT1011 Engineering Materials

Introduction-I

Dr Mudith Karunaratne

Department of Materials Engineering
Sri Lanka Institute of Information Technology

September 23, 2026

Engineering Materials

Our modern lifestyle is influenced by materials.



Figure: Ten selected materials that have helped shape our modern society.

Picture credit: pbs.org

Advancement of societies is dependent on our ability to produce **materials**



Technology is associated with the availability of suitable materials



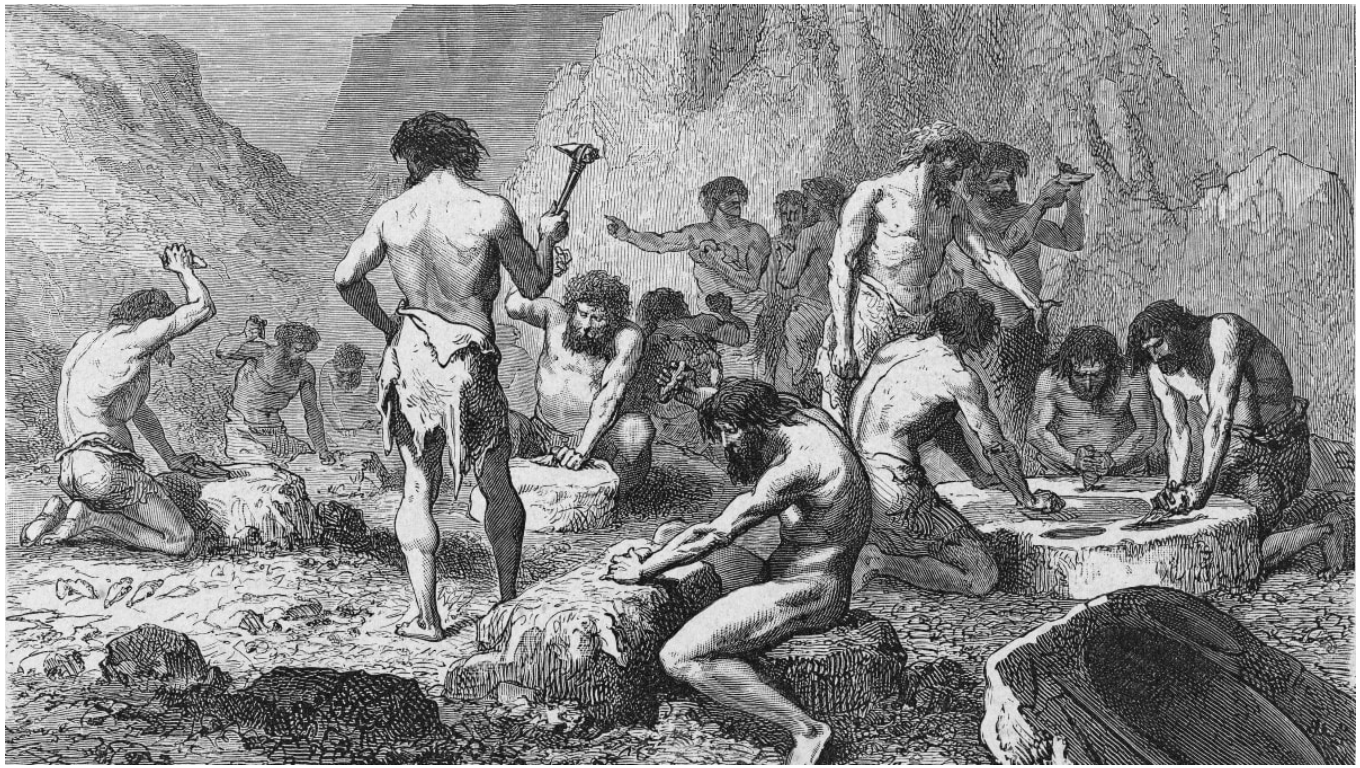
- A car would contain more than 800 different materials.
- Cars would not have been possible without the availability of inexpensive **steel**.



Your smartphone relies on components that are made from semiconducting and various sensor materials.



The earliest humans had access only to a few materials natural materials such as stone, wood, clay and skins.



Over time, the range of materials available has increased immensely.



Today, for a particular application, there is generally a range of materials available to choose from.



Figure: Mugs made of various materials.

Selection process involves deciding from a given set of materials, the one best suited for an application based on material properties.



Figure: Mugs made of various materials.